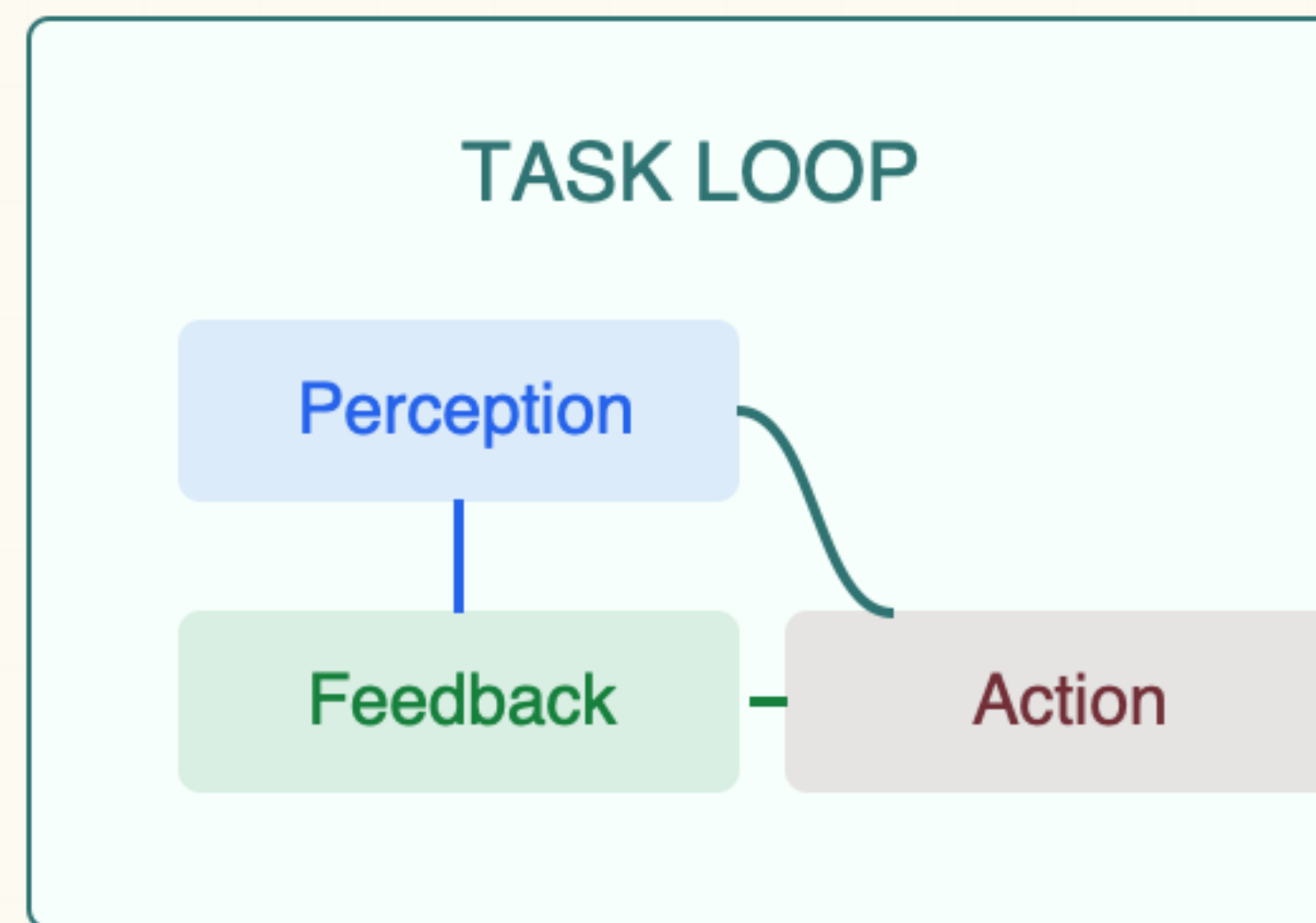


Agent AI Is Not Just a Chatbot

A grill-based reading of "Agent AI: Surveying the Horizons of Multimodal Interaction"

- perception
- environment
- action
- feedback



The Paper Gives a Map, Not a Finished Machine

Use it as a conceptual vocabulary, not a 2026 state-of-the-art catalog.

- January 2024 survey
- Moves beyond "LLM + tools + memory"
- Centers multimodal perception, grounded environments, action, and feedback

READING HEURISTIC

foundation model
+ multimodal perception
+ grounded environment
+ action prediction
+ feedback loop
+ evaluation in context

Multimodal Is Not Automatically Agentic

The difference is the task loop, not just the media

- Multimodal: understands text, image, video, audio, signals
- Agentic: chooses actions from perception, state, goal, feedback
- A caption is not yet an agent

Multimodal

Understands different sources of information: text, image, video, audio, screen state.

Agentic

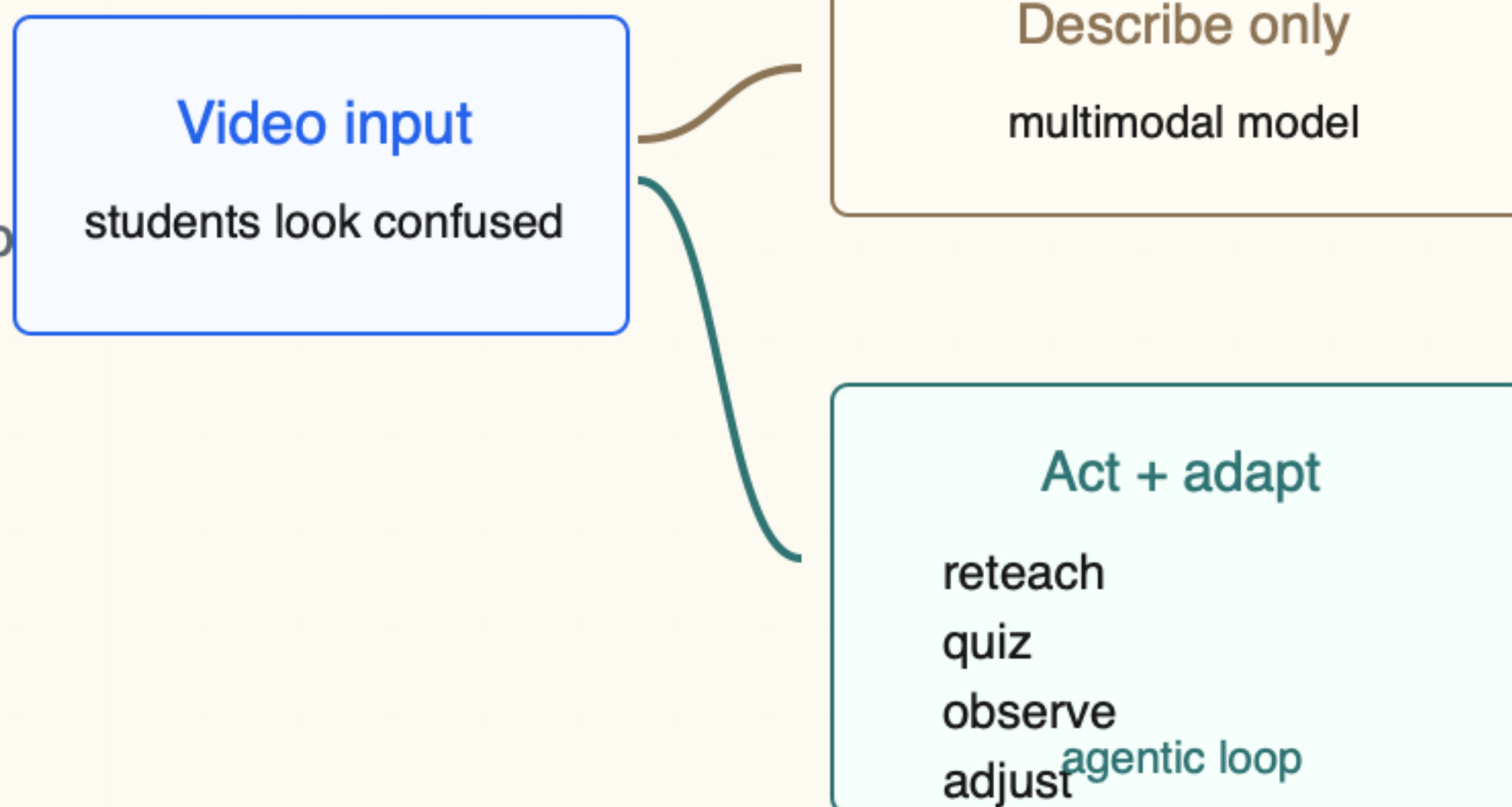
Uses perception plus state, goal, action, and feedback to decide what happens next.

More media is not the same as agency

The Classroom Video Test

Saying "three students look confused" is perception full agent loop.

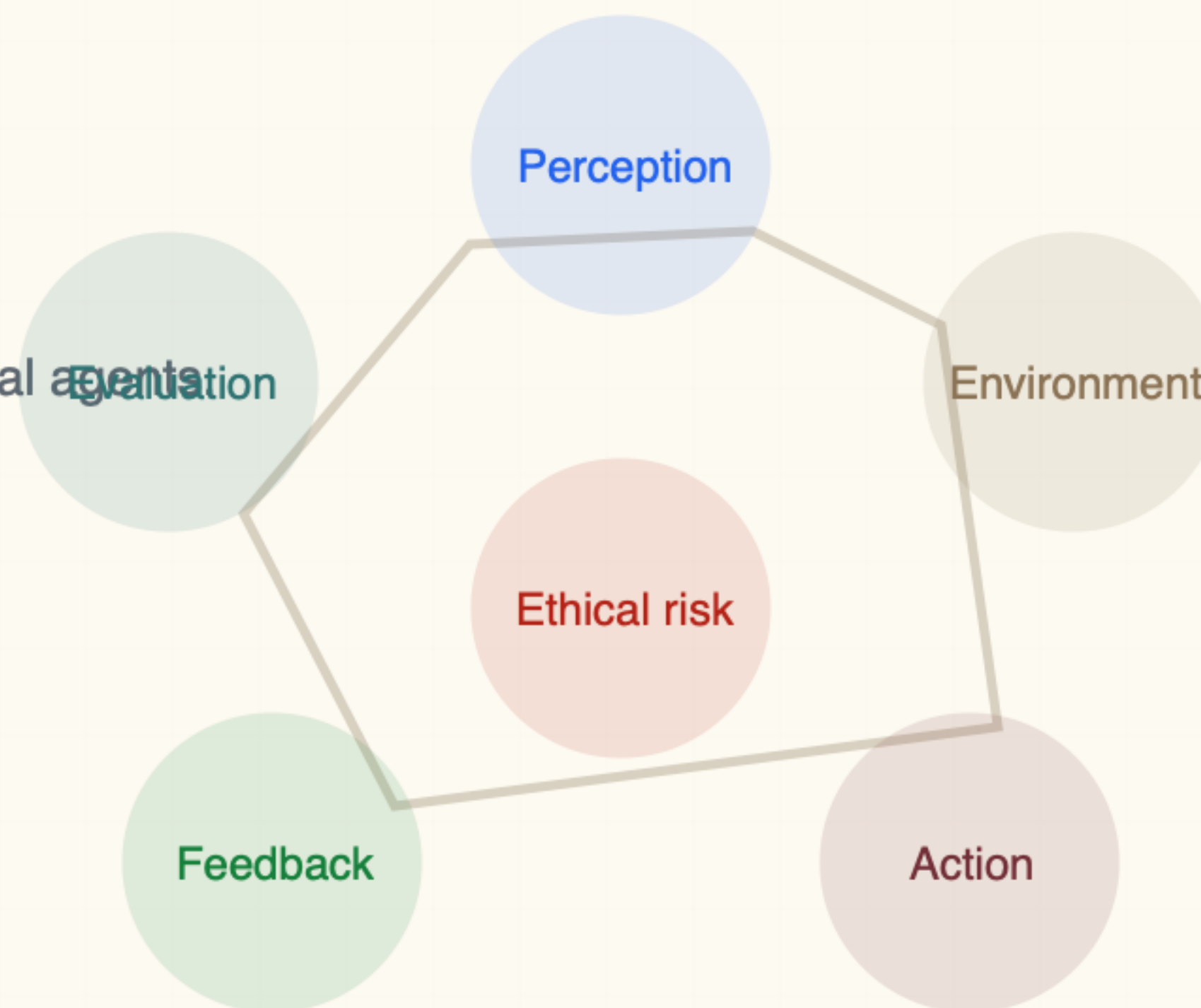
- Describe confusion: multimodal perception
- Reteach + quiz + observe: agentic loop
- Description alone is not enough



The Agent Loop Has Six Questions

This is the reusable reading checklist for multimodal agent evaluation

- Perception
- Environment
- Action
- Feedback
- Evaluation
- Ethical risk



When Perception Becomes Action, Ethics Gets Harder

Wrong perception can become a classroom consequence

Sensitive signal

screen / face /
voice / grades

AI interpretation

confused /
disengaged /
suspicious

Classroom consequence

reteach / flag /
grade / report

- Privacy: screens, faces, grades, voice, teacher material
- Fairness: posture, accent, disability, lighting, camera quality
- Accountability: who authorized the action and who corrects it?

Wrong perception can become wrong action

AI Helps Learning When Students Still Own Thinking

The boundary is observable student understanding

- Help: explain, question, plan, revise, verify
- Replace: generate final answer without understanding
- Test: explain, verify, adapt, defend

Helps learning

explain
question
plan
revise
verify

student can defend

Replaces learning

AI generates the main
answer; student submits
without understanding.

student cannot defend

Can they explain, verify, adapt, and defend?

"AI Allowed" and "AI Banned" Are Both Too Weak

Agentic systems need operational boundaries,

- Approved tools
- Allowed and forbidden uses
- Privacy-safe disclosure
- Explicit exam and authorship boundaries

ALLOW

BAN

Policy matrix

- ☐ Approved tools
- ☐ Allowed uses
- ☐ Forbidden uses
- ☐ Disclosure
- ☐ Exam boundary

Classroom Agents Need Boundaries Before De

Consent, audit, authority, and material protection
explicit.



Consent before recording or analyzing people



Logs and audits for recommendations and actions



Human authority for grading, discipline,
accommodations, appeals



No unauthorized upload of course or peer
material

From Advice to Task Loop

The upgrade is AI participating in perception, action, and feedback.

